Binomial Theorem: Short Notes

How to use these notes: read the Formulas box, follow the Steps and Key points, and avoid the Watch out points.

1. Binomial theorem and basics

Formulas

$$(x+y)^n = \sum_{r=0}^{n} {}^nC_r\, x^{n-r} y^r, \qquad (1 \pm x)^n = \sum_{r=0}^{n} (\pm 1)^r\, {}^nC_r x^r$$

$${}^nC_r = {}^nC_{n-r}, \qquad {}^nC_r + {}^nC_{r-1} = {}^{n+1}C_r, \qquad \sum {}^nC_r = 2^n$$

KEY POINTS

- Monomial 1 term, binomial 2, trinomial 3, multinomial more than 2; Pascal's triangle gives the coefficients row by row
- $n+1$ terms; powers of x and y add up to n in every term
- Binomial coefficient $= {}^nC_r$ only; coefficient includes the numbers attached to x, y
- Sum of coefficients: put variable $= 1$; sum of binomial coefficients $= 2^n$

2. General term, rational terms, term from the end

Formulas

$$T_{r+1} = {}^nC_r\, x^{n-r} y^r, \qquad k\text{th term from the end} = T_{n-k+2}$$

STEPS

1. Write T_{r+1} and collect the power of x
2. Equate it to the required power (0 for the independent term) and solve for r
3. Accept r only if it is an integer with $0 \le r \le n$, else the coefficient is 0
4. Rational terms: make every fractional power an integer; r is a multiple of the lcm of the denominators

KEY POINTS

- Ratio of consecutive terms $\dfrac{T_{r+1}}{T_r} = \dfrac{n-r+1}{r} \cdot \dfrac{y}{x}$; use it for three consecutive coefficients in a ratio or AP
- ${}^nC_x = {}^nC_y \Rightarrow x = y$ or $x + y = n$
- Term with a parameter ($\sin\theta$ etc.): write the term, then use its bounds for max / min

Watch out

Do not forget the sign of y and powers of numbers like $(2x)^{n-r}$

3. Middle term and products of expansions

Formulas

$$n \text{ even}: \; T_{\frac{n}{2}+1}; \qquad n \text{ odd}: \; T_{\frac{n+1}{2}},\; T_{\frac{n+3}{2}}$$

KEY POINTS

- Middle term has the greatest binomial coefficient (${}^nC_{n/2}$ or ${}^nC_{(n\pm1)/2}$)
- Product of expansions: pair general terms whose powers add up to the required power and add the products of coefficients
- Factorise or sum a GP first when it makes the factors binomials

4. Numerically greatest term (NGT)

Formulas

$$p = \frac{n+1}{1 + \left|\frac{a}{b}\right|} \quad \text{for } (a+b)^n$$

STEPS

1. Compute p (use $|a/b|$)
2. p not an integer: NGT $= T_{[p]+1}$

3. p an integer: T_p and T_{p+1} are equal and both NGT
4. Position T_k given: $k-1<p<k$ (single NGT) or $p=k$ (two NGT); solve for x or n

Watch out

Take modulus: NGT concerns $|T_r|$, signs are ignored

5. Remainders, divisibility, last digits

Formulas

$$a^n - b^n: \text{ divisible by } a-b\ \forall n;\ \text{by } a+b \text{ if } n \text{ even}$$
$$a^n + b^n: \text{ divisible by } a+b \text{ if } n \text{ odd}$$

STEPS

1. Write the base as (multiple of divisor) $\pm$ small number
2. If the base is small, first raise it to a power that is close to a multiple of the divisor
3. Expand; all terms except the last (or first two for d^2) are divisible by the divisor
4. Negative remainder: add the divisor

KEY POINTS

- Last 1, 2, 3 digits = remainder mod $10, 100, 1000$; write leading zeros
- $(1+kd)^n \equiv 1+nkd \pmod{d^2}$

6. Integral and fractional part

Formulas

$$R=(a+\sqrt{b})^n = I+f, \quad G=(a-\sqrt{b})^n, \quad R+G\in\mathbb{Z}, \quad RG=(a^2-b)^n$$

STEPS

1. Take the conjugate G and check whether $0<G<1$ or $-1<G<0$
2. $0<G<1$: $f+G=1$, $I=R+G-1$; $-1<G<0$: $f=-G$, $I=R+G$
3. Parity of I from the even / odd integer $R+G$

KEY POINTS

- $(a+\sqrt{b})^n \pm (a-\sqrt{b})^n$ keeps only even (or odd) r and is twice the surviving sum

7. Multinomial theorem

Formulas

$$\frac{n!}{r_1!\,r_2!\cdots r_k!}\,x_1^{r_1}\cdots x_k^{r_k}, \quad \sum r_i = n$$
$$\text{number of dissimilar terms} = {}^{n+k-1}C_{k-1}$$

STEPS

1. Write the general term and impose $\sum r_i = n$
2. Impose the power of x; list all non-negative integer solutions
3. Add the corresponding coefficients

KEY POINTS

- Factorise the base (e.g. $(x-2)(x+1)$) before expanding
- Distinct terms in a sum of expansions: take the union of the powers present

8. Summation of series, types 1 and 2

Formulas

$$\sum_{r=0}^{n} {}^nC_r\, x^{n-r}y^r = (x+y)^n, \qquad \sum_{k=r}^{m} {}^kC_r = {}^{m+1}C_{r+1}$$

STEPS

1. Type 1 (lower suffix varies, powers of r): read x, y, n from the general term
2. Check the limits: if the sum starts at $r = 1$ subtract the $r = 0$ term
3. Type 2 (upper suffix varies): combine from the first two terms with ${}^{n}C_r + {}^{n}C_{r-1} = {}^{n+1}C_r$
4. Write 1 as ${}^{k}C_k$ when the first term is a 1

Watch out

Sign $(-1)^r$ makes y negative; check powers like $4^r = (2^2)^r$

9. Summation type 3 (sign change, even / odd, removing r)

Formulas

$$a_0 + a_2 + \cdots = \frac{f(1) + f(-1)}{2}, \qquad a_1 + a_3 + \cdots = \frac{f(1) - f(-1)}{2}$$

$$r\,{}^{n}C_r = n\,{}^{n-1}C_{r-1}, \qquad \sum r\,{}^{n}C_r = n2^{n-1}, \qquad \sum r^2\,{}^{n}C_r = n(n+1)2^{n-2}$$

$$\sum_{r=0}^{k} (-1)^r\,{}^{n}C_r = (-1)^k\,{}^{n-1}C_k$$

STEPS

1. Alternating signs: put $x = -1$ (full sum $= 0$; partial sum by the formula)
2. Factor r or $r(r-1)$ in the term: remove it with $r\,{}^{n}C_r = n\,{}^{n-1}C_{r-1}$
3. For r^2 write $r^2 = r(r-1) + r$
4. Gauss technique: reverse the sum, add, $2S = \sum(t_r + t_{n-r})$

10. Summation type 4 (differentiation, integration)

Formulas

$$\sum_{r=0}^{n} \frac{{}^{n}C_r}{r+1} = \frac{2^{n+1} - 1}{n+1}, \qquad \frac{{}^{n}C_r}{r+1} = \frac{{}^{n+1}C_{r+1}}{n+1}$$

STEPS

1. Factor r, $r-1$ in the numerator (no denominator): differentiate $(1+x)^n$, then put $x = 1$ or -1
2. Denominator $r+1$, $r+2$: integrate $(1+x)^n$ (or $x^k(1+x)^n$) between 0 and 1
3. Or convert to type 1 by absorbing the denominator in ${}^{n+1}C_{r+1}$

11. Summation type 5 (product of two binomial coefficients)

Formulas

$$\sum_{r} {}^{n}C_r\,{}^{m}C_{k-r} = {}^{n+m}C_k, \qquad \sum_{r} {}^{n}C_r\,{}^{m}C_{r+k} = {}^{n+m}C_{n+k}$$

$$\sum({}^{n}C_r)^2 = {}^{2n}C_n$$

STEPS

1. Identify: lower suffixes with constant sum (sum pattern) or constant difference
2. Objective: use the formula directly
3. Subjective: multiply $(1+x)^n$ and $(1+x)^m$ (one written in reverse powers for the difference pattern) and compare the coefficient of x^k

12. Miscellaneous series questions

Formulas

$$(1+x)^m + x(1+x)^{m-1} + \cdots + x^m = (1+x)^{m+1} - x^{m+1}$$

$$(1+x)^n = C_0 + \cdots + C_n x^n, \quad (x+1)^n = C_0 x^n + \cdots + C_n$$

KEY POINTS

- Look for a GP in the terms (ratio $\frac{x}{1+x}$ or $1+x$) and sum it first
- Use ${}^nC_r = {}^nC_{n-r}$ for pairing and for half of a symmetric sum
- Reverse the order of powers to match terms when multiplying expansions

13. Binomial theorem for any index

Formulas

$$(1+x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \ldots, \qquad (1-x)^{-n} = 1 + nx + \frac{n(n+1)}{2!}x^2 + \ldots$$

$$\text{coeff. of } x^r \text{ in } (1-x)^{-n} = {}^{n+r-1}C_r \quad (|x| < 1)$$

STEPS

1. Take the constant out: $(a+bx)^n = a^n\left(1+\frac{b}{a}x\right)^n$; valid for $\left|\frac{b}{a}x\right| < 1$
2. Product with $(1-x)^{-n}$: expand the polynomial factor and add term-wise coefficients
3. Drop $(1-x^k)$ when k exceeds the required power; cancel common factors

Watch out

A non-natural index gives an infinite series, valid only for $|x| < 1$

14. Series of e^x and $\ln(1+x)$

Formulas

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}, \qquad \ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \ldots \quad (-1 < x \leq 1)$$

STEPS

1. Write the general term as (polynomial in n)/$n!$
2. Break into $\frac{n(n-1)}{n!} = \frac{1}{(n-2)!}$, $\frac{n}{n!} = \frac{1}{(n-1)!}$ and sum each as e
3. For ln series match the sign and the denominator n with x^n/n

15. Approximation, parameters and special tricks

KEY POINTS

- Small $|x|$: $(1+x)^n \approx 1+nx$; for bounds keep the first few positive terms
- $(x+a)^n + (x-a)^n$: only even r survive, so degree and number of terms follow from the last surviving term
- Sum of coefficients of positive powers = total − constant term − negative-power terms

16. Common mistakes checklist

STEPS

1. Index of T_{r+1}: the $(r+1)$th term has ${}^nC_r$, not ${}^nC_{r+1}$
2. Range of r ($0 \leq r \leq n$ and integer) before accepting a root
3. Coefficient vs binomial coefficient; sign of y
4. NGT: use moduli; integer p gives two equal terms
5. Remainder: convert negative remainders; last digits with leading zeros
6. Sums: limits of r ($r = 0$ term) and the factor 2^n vs 2^{n-1}
7. Any index: check $|x| < 1$ and the factor taken out